Gangshan Jing

School of Automation
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Research Interest

Networked Systems: graph-based approaches for formation control and network localization

Reinforcement Learning: scalability and optimality of multi-agent reinforcement learning

Robotics Motion Planning: cooperative manipulation and decision making in dynamic environments

Academic Experience

School of Automation, Chongqing University Chongqing, China
Professor of Control Science and Engineering Dec. 2021-Present

Department of Electrical and Computer Engineering, North Carolina State University Raleigh, U.S.
Postdoctoral Researcher. Advisor : Aranya Chakrabortty Sept. 2019 – Oct. 2021
- Conducted research on reinforcement learning in control of large-scale systems

Department of Mechanical and Aerospace Engineering, The Ohio State University Columbus, U.S.
Postdoctoral Researcher. Advisor: Ran Dai Oct. 2018 - Sept. 2019
- Conducted research on motion dynamics of spacecraft during Mars entry and angle-based sensor network localization

Department of Applied Mathematics, Hong Kong Polytechnic University Hong Kong, China
Research Assistant. Mentors: Dr. Guofeng Zhang and Dr. Joseph Lee Nov. 2017 - Jan. 2018
- Conducted research on angle rigidity theory and angle-based formation control
- Assisted in writing a proposal for Hong Kong RGC: "Novel graph rigidity theory and application to distributed formation control"

Department of Applied Mathematics, Hong Kong Polytechnic University Hong Kong, China
Research Assistant. Mentors: Dr. Guofeng Zhang and Dr. Joseph Lee Dec. 2016 - May. 2017
- Conducted research on weak rigidity theory and its application to formation control

Education

Xidian University, School of Mechano-Electronic Engineering Xi'an, China
Ph.D. in Control Theory and Control Engineering Sept. 2012 - Sept. 2018
Dissertation title: New Graphical Approaches to Multi-Agent Formation Control
Advisor: Prof. Long Wang
Area of study: Distributed Control of Networked Systems

Ningxia University, School of Mathematics and Computer Science Yinchuan, China
B.S. in Mathematics and Applied Mathematics, GPA Ranking: 1/160. Sept. 2008 - July. 2012
Thesis: Consensus of Multi-Agent Systems with Time-Delays

Funded Projects

Start-up Fund for HongShen Young Scholars Jan. 2022 – Dec. 2024
Chongqing University

Graph-Based Distributed Reinforcement Learning and Control of Large-Scale Network Systems
NSFC Jan. 2023 – Dec. 2025

Reinforcement Learning and Control of Multi-Agent Systems
NSFC Jan. 2023 – Dec. 2025

Skill Learning and Autonomous Development of Dexterous Manipulator
Ministry of Science and Technology Nov. 2022 – Oct. 2025

Model-Free Optimal Distributed Control of Large-Scale Network Systems
NSF of Chongqing July 2023 – June 2025

Non-Uniform Scaling Formation Control Under Dynamic Environments
NSFC Jan. 2026 – Dec. 2029

Theories and Key Technologies of Intelligent Game and Collaborative Control for Complex Dynamic Systems
NSFC Jan. 2026 – Dec. 2030

Algebraic-Combinatorial Approaches for Optimization and Control of Network Systems
Ministry of Science and Technology Dec. 2025 – Nov. 2030

Teaching Experience

Chongqing University

Advanced Control Theory and Applications Spring 2025
Advanced Control Theory and Applications Spring 2024
Foundations of Control Theory Fall 2023

Xidian University

Teaching Assistant for **Optimal Control Theory** Spring 2014
Teaching Assistant for **Linear Algebra** Spring 2013

Publications

Journal Articles

[J1]. J. Tang, P. Zhou, H. Bai, and **G. Jing***, "GRA: Graph-based Reward Aggregation for cooperative multi-agent reinforcement learning," ***Journal of Automation and Intelligence***. 2025, doi: 10.1016/j.jai.2025.10.006.

[J2]. B. Peng, B. Liu, **G. Jing**, Z. Ding, Y. Song, "Intertemporal optimization of formation control for nonlinear networks with multiple constraints: A reach-avoid game approach," ***IEEE Transactions on Automatic Control***. 2025, doi: 10.1109/TAC.2025.3621752.

[J3]. G. He, **G. Jing***, and Y. Song, “Global stabilization of similar formation via edge-based clique addition,” ***Automatica***, vol. 173, p. 112084, Mar. 2025, doi: 10.1016/j.automatica.2024.112084.

[J4]. K. Li, Z. Shen, **G. Jing***, and Y. Song, “Angle-constrained formation control under directed non-triangulated sensing graphs,” ***Automatica***, vol. 163, p. 111565, May 2024, doi: 10.1016/j.automatica.2024.111565.

[J5]. **G. Jing**, C. Wan, R. Dai, and M. Mesbahi, “Design and transformation control of triangulated origami tessellation: A network perspective,” ***IEEE Transactions on Network Science and Engineering***, vol. 11, no. 1, pp. 635–647, Jan. 2024, doi: 10.1109/TNSE.2023.3303260.

[J6]. **G. Jing***, H. Bai, J. George, A. Chakrabortty, and P. K. Sharma, “Distributed Multiagent Reinforcement Learning Based on Graph-Induced Local Value Functions,” ***IEEE Transactions on Automatic Control***, vol. 69, no. 10, pp. 6636–6651, Oct. 2024, doi: 10.1109/TAC.2024.3375248.

[J7]. **G. Jing***, H. Bai, J. George, A. Chakrabortty, and P. K. Sharma, “Asynchronous Distributed Reinforcement Learning for LQR Control via Zeroth-Order Block Coordinate Descent,” ***IEEE Transactions on Automatic Control***, vol. 69, no. 11, pp. 7524–7539, Nov. 2024, doi: 10.1109/TAC.2024.3386061.

[J8]. C. Wan, **G. Jing**, R. Dai, and J. R. Rea, “Fuel-Optimal Guidance for End-to-End Human-Mars Entry, Powered-Descent, and Landing Mission,” ***IEEE Transactions on Aerospace Electronic Systems***, vol. 58, no. 4, pp. 2837–2854, Aug. 2022, doi: 10.1109/TAES.2022.3141325.

[J9]. **G. Jing**, C. Wan, and R. Dai, “Angle-Based Sensor Network Localization,” ***IEEE Transactions on Automatic Control***, vol. 67, no. 2, pp. 840–855, Feb. 2022, doi: 10.1109/TAC.2021.3061980.

[J10]. **G. Jing**, H. Bai, J. George, A. Chakrabortty, and P. K. Sharma, “Learning Distributed Stabilizing Controllers for Multi-Agent Systems,” ***IEEE Control System Letters***, vol. 6, pp. 301–306, 2022, doi: 10.1109/LCSYS.2021.3072007.

[J11]. **G. Jing**, H. Bai, J. George, and A. Chakrabortty, “Model-Free Optimal Control of Linear Multiagent Systems via Decomposition and Hierarchical Approximation,” ***IEEE Transactions on Control of Network Systems***, vol. 8, no. 3, pp. 1069–1081, Sep. 2021, doi: 10.1109/TCNS.2021.3074256.

[J12]. C. Wan, **G. Jing**, S. You, and R. Dai, “Sensor Network Localization via Alternating Rank Minimization Algorithms,” ***IEEE Transactions on Control of Network Systems***, vol. 7, no. 2, pp. 1040–1051, Jun. 2020, doi: 10.1109/TCNS.2019.2926775.

[J13]. K. Lu, **G. Jing**, and L. Wang, “Online Distributed Optimization With Strongly Pseudoconvex-Sum Cost Functions,” ***IEEE Transactions on Automatic Control***, vol. 65, no. 1, pp. 426–433, Jan. 2020, doi: 10.1109/TAC.2019.2915745.

[J14]. K. Lu, **G. Jing**, and L. Wang, “Distributed algorithms for solving the convex feasibility problems,” ***Science China Information Science***, vol. 63, no. 8, p. 189201, Aug. 2020, doi: 10.1007/s11432-018-9682-4.

[J15]. **G. Jing** and L. Wang, “Multiagent Flocking With Angle-Based Formation Shape Control,” ***IEEE Transactions on Automatic Control***, vol. 65, no. 2, pp. 817–823, Feb. 2020, doi: 10.1109/TAC.2019.2917143.

[J16]. **G. Jing**, H. Bai, J. George, and A. Chakrabortty, “Model-Free Reinforcement Learning of Minimal-Cost Variance Control,” ***IEEE Control Systems Letters***, vol. 4, no. 4, pp. 916–921, Oct. 2020, doi: 10.1109/LCSYS.2020.2995547.

[J17]. K. Lu, **G. Jing**, and L. Wang, “Distributed Algorithms for Searching Generalized Nash Equilibrium of Noncooperative Games,” ***IEEE Transactions on Cybernetics.***, vol. 49, no. 6, pp. 2362–2371, Jun. 2019, doi: 10.1109/TCYB.2018.2828118.

[J18]. K. Lu, **G. Jing**, and L. Wang, “A distributed algorithm for solving mixed equilibrium problems,” ***Automatica***, vol. 105, pp. 246–253, Jul. 2019, doi: 10.1016/j.automatica.2019.03.015.

[J19]. **G. Jing**, G. Zhang, H. W. J. Lee, and L. Wang, "Angle-based Shape Determination Theory of Planar Graphs with Application to Formation Stabilization," ***Automatica***, 105, pp.117-129, Jul. 2019, doi: 10.1016/j.automatica.2019.03.026.

[J20]. **G. Jing** and L. Wang, "Finite-Time Coordination Under State-Dependent Communication Graphs With Inherent Links," ***IEEE Transactions on Circuits and Systems II: Express Briefs***, vol. 66, no. 6, pp. 968–972, Jun. 2019, doi: 10.1109/TCSII.2018.2866052.

[J21]. K. Lu, **G. Jing**, and L. Wang, "Distributed Algorithm for Solving Convex Inequalities," ***IEEE Transactions on Automatic Control***, vol. 63, no. 8, pp. 2670–2677, Aug. 2018, doi: 10.1109/TAC.2017.2771140.

[J22]. **G. Jing**, G. Zhang, H. W. Joseph Lee, and L. Wang, "Weak Rigidity Theory and Its Application to Formation Stabilization," ***SIAM Journal on Control and Optimization***, vol. 56, no. 3, pp. 2248–2273, Jan. 2018, doi: 10.1137/17M1122049.

[J23]. **G. Jing,** Y. Zheng, and L. Wang, "Consensus of Multiagent Systems With Distance-Dependent Communication Networks," ***IEEE Transactions on Neural Networks and Learning Systems***, vol. 28, no. 11, pp. 2712–2726, Nov. 2017, doi: 10.1109/TNNLS.2016.2598355.

[J24]. **G. Jing**, Y. Zheng, and L. Wang, "Flocking of multi-agent systems with multiple groups," ***International Journal of Control***, vol. 87, no. 12, pp. 2573–2582, Dec. 2014, doi: 10.1080/00207179.2014.935485.

Conference Papers

[C1]. J. Huang, **G. Jing**, "On the Equivalence between Signed Angle Rigidity and Bearing Rigidity," in ***2025 IEEE Conference on Decision and Control (CDC)***, accepted.

[C2]. **G. Jing**, H. Bai, J. George, A. Chakrabortty, and Piyush. K. Sharma, "Distributed Cooperative Multi-Agent Reinforcement Learning with Directed Coordination Graph," in 2022 ***American Control Conference (ACC)***, Jun. 2022, pp. 3273–3278. doi: 10.23919/ACC53348.2022.9867152.

[C3]. C. Wan, C. Pei, R. Dai, **G. Jing**, and J. R. Rea, "Six-Dimensional Atmosphere Entry Guidance based on Dual Quaternion," in ***AIAA Scitech 2021 Forum***, VIRTUAL EVENT: American Institute of Aeronautics and Astronautics, Jan. 2021. doi: 10.2514/6.2021-0507.

[C4]. C. Wan, **G. Jing**, R. Dai, and R. Zhao, "Local Shape-Preserving Formation Maneuver Control of Multi-agent Systems: From 2D to 3D," in 2021 60th ***IEEE Conference on Decision and Control (CDC)***, Dec. 2021, pp. 6251–6257. doi: 10.1109/CDC45484.2021.9683637.

[C5]. **G. Jing**, H. Bai, J. George, and A. Chakrabortty, "Decomposability and Parallel Computation of Multi-Agent LQR," in 2021 ***American Control Conference (ACC)***, May 2021, pp. 4527–4532. doi: 10.23919/ACC50511.2021.9483338.

[C6]. C. Wan, S. You, **G. Jing**, and R. Dai, "A Distributed Algorithm for Sensor Network Localization with Limited Measurements of Relative Distance," in 2019 ***American Control Conference (ACC)***, Jul. 2019, pp. 4677–4682. doi: 10.23919/ACC.2019.8814418.

[C7]. **G. Jing**, C. Wan, and R. Dai, "Angle Fixability and Angle-Based Sensor Network Localization," in 2019 IEEE 58th ***Conference on Decision and Control (CDC)***, Dec. 2019, pp. 7899–7904. doi: 10.1109/CDC40024.2019.9029907.

Invited Talks

Invited Talk at Department of Mechanical Engineering, City University of Hong Kong, Hong Kong, 26 November, 2025
Oganizer: Changhuang Wan
Title: *Graph Rigidity Theory and Its Applications in Multi-Agent Coordination*

Keynote Talk in The Second Graduate Interdisciplinary Forum of the Department of Mathematics and Information Science, Ningxia University, Yinchuan, 15 November, 2025
Host: Jingying Ma
Title: *Graph Rigidity Theory and Its Applications in Multi-Agent Coordination*

Invited Talk at Shaanxi University of Science and Technology (online), 10 November, 2025
Organizer: Zhou He
Title: *Graph Rigidity Theory and Its Applications in Multi-Agent Coordination*

Keynote Talk in The Workshop on Control, Optimization and Game Theory of Multi-Agent Systems, Northwestern Polytechnical University, Xi’an, 7 November, 2025
Host: Yanmei Kang (Xi'an Jiaotong University)
Title: *Graph Rigidity Theory and Its Applications in Multi-Agent Coordination*

Keynote Talk in The 2025 Academic Annual Conference of the Automation and Instrumentation Association of the Three Provinces and One Municipality in Southwest China, Guiyang, China, June, 2025
Title: *Graph Rigidity Theory and its Application to Multi-Agent Coordination Problems*

Invited Talk at Beijing Institute for General Artificial Intelligence, January, 2025
Organizer: Xue Feng
Title: *Graph-Theoretic Approaches in Network Systems*

Keynote Talk in Workshop on AI for Control Science and Engineering, School of Automation Engineering, University of Electronic Science and Technology of China, Chengdu, China, March, 2025
Title: *Distributed Reinforcement Learning for Cooperative Optimal Control of Multi-Agent Systems*

Invited Talk in the Workshop on Control, Optimization and Game of Multi-Agent Systems, Northwestern Polytechnical University, December, 2024
Organizer: Shenggui Zhang
Title: *Angle-Based Sensor Network Localization*

Keynote Talk in 2024 7th International Conference on Robotics, Control and Automation Engineering (RCAE), Wuhu, China, October, 2024,
Title: *Data-Driven Control for Multi-Agent Systems*

Invited Talk at Department of Automation, Shanghai Jiaotong University, October, 2024
Organizer: Qi Su
Title: *Data-Driven Control and Reinforcement Learning*

Invited Talk at School of Automation, Beijing Institute of Technology, Sept., 2024
Organizer: Yuan Zhang
Title: *A Graph-Theoretic Approach to Distributed Cooperative Multi-Agent Reinforcement Learning*

Invited Speaker in CCF Young Computer Scientists & Engineers Forum of “How to integrate large language models in cars”, Southwest University, Chongqing, August, 2024

Invited Speaker at Roundtable Dialogue in 2024 Conference on Intelligent Networked Things (CINT), May, 2024
Organizer: Liyuanjun Lai
Title: *Cooperations in Network Systems*

Invited Talk in the forum of AI-Driven Control Systems: Reliability and Applications, 2023 China Automation Congress, Chongqing, China, November, 2023
Organizer: Yongduan Song
Title: *A Graph-Theoretic Approach to Distributed Cooperative Multi-Agent Reinforcement Learning*

Invited Talk in Xi'an University of Science and Technology, June 14, 2023
Organizer: Xuya Cong
Title: *Multi-Agent Reinforcement Learning with Graph-Induced Local Value Functions*

Invited Talk in Center for Systems and Control, College of Engineering, Peking University, December, 2021
Organizer: Aming Li
Title: *A General Graph-Theoretic Distributed Framework for Multi-Agent Reinforcement Learning*

Invited Talk in Center for Systems and Control, College of Engineering, Peking University, November, 2021
Organizer: Aming Li
Title: *Fast Reinforcement Learning for LQR Control via Decomposition and Hierarchical Approximation*

Invited Talk for U.S. Army Research Laboratory, October, 2021
Organizer: Jemin George
Title: *Distributed Reinforcement Learning and Optimal Control of Multi-Agent Systems*

Invited Talk for Southeast University, July, 2021
Organizer: Guanghui Wen
Title: *Fast Reinforcement Learning for LQR Control via Decomposition and Hierarchical Approximation*

Invited Talk in Control 3.0 Forum, June, 2020
Organizer: Long Wang
Title: *Angle-Based Sensor Network Localization*

Invited Talk at Hangzhou Dianzi University, Hangzhou, China, April, 2018
Organizer: Brian D.O. Anderson
Title: *Angle-Based Formation Control*

Synergistic Service

Associate Editor, Journal of Automation and Intelligence

Organizing Committee Chair, 2024 Workshop on Intelligent Decision Making, Chongqing, China

Publication Chair, 2025 Joint International Conference on Automation-Intelligence-Safety & International Symposium on Autonomous Systems (ICAIS&ISAS), Xi'an, China

Invited Session Chair, 2025 Chinese Control Conference, Chongqing, China

IEEE member, CAA member

Reviewer for IEEE Transactions on Automatic Control, IEEE Transactions on Pattern Analysis and Machine Intelligence, SIAM Journal on Control and Optimization, SIAM Journal on Matrix Analysis and Applications, Automatica, Systems & Control Letters, IEEE Control Systems Letters, IEEE Robotics and Automation Letters, IEEE Transactions on Signal Processing, IEEE Transactions on Control of Network Systems, IEEE Transactions on Control Systems Technology, IEEE Transactions on Power Systems, IEEE Transactions on Aerospace and Electronic Systems, IEEE Transactions on Cybernetics, IEEE Transactions on Neural Networks and Learning Systems, IEEE Transactions on Industrial Electronics, IEEE Transactions on Systems, Man and Cybernetics: Systems, IEEE Transactions on Circuits and Systems II: Express Briefs, Applied Mathematics and Computation, Unmanned Systems, Mathematical Reviews (MathSciNet), IEEE Conference on Decision and Control, IEEE International Conference on Control & Automation, American Control Conference, European Control Conference, Chinese Control Conference

References

- Yongduan Song (IEEE Fellow)
  Professor, School of Automation
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- Aranya Chakrabortty (IEEE Fellow)
  Professor, Department of Electrical and Computer Engineering
  FREEDM Systems Center
  North Carolina State University, U.S.
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- Ran Dai
  Professor, School of Aeronautics and Astronautics
  Purdue University, U.S.
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- Long Wang
  Professor, Department of Industrial Engineering and Management, College of Engineering
  Director of Center for Systems and Control
  Peking University, China
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- Guofeng Zhang
  Professor, Department of Applied Mathematics
  Hong Kong Polytechnic University, China
  E-mail address: guofeng.zhang@polyu.edu.hk

- Jemin George
  Research Scientist, U.S. Army Research Laboratory
  Email address: jemin.george.civ@mail.mil

- He Bai
  Assistant Professor, Department of Mechanical and Aerospace Engineering
  Oklahoma State University, U.S.
  E-mail address: he.bai@okstate.edu